Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



ANANTA: The Voice-First Socratic AI Tutor

Where Every Question Opens Infinity · Eliminating the 'Correct Answer Trap' with Verbatim Proof

Team Name: Ananta | Track: Media, Content & Digital Experiences / Education

Problem Statement: Standard quiz apps grade only final answers, hiding broken logic when students guess correctly. Ananta delivers 1:1 Socratic voice tutoring that diagnoses misconceptions at the point of thought, pivots dynamically via an SVG chalkboard, and proves mastery with verbatim evidence.

Live Prototype: https://ananta-ivczdbgbja-as.a.run.app/ · GitHub: sur2rock1/AI_Builder_cup_2026

Problem Statement: The 'Correct Answer Trap' in K-12

Why 1:1 tutoring is vital, and why conventional quiz apps fail our children

Submission Template | AI Builder Cup

THE 'CORRECT ANSWER TRAP' IN K-12 EDUCATION

Why standard quizzes fail: A child can get the right answer with completely broken logic.

THE ACTUAL STUDENT EXCHANGE

Tutor Question:
"Which is greater: 0.50*or*0.7?"

Child's Response:
"$0.50, because 50 is bigger than 7!"

Pedagogic Diagnosis:
The child treats decimals as whole numbers.
They see '50' and '7' rather than tenths.

STANDARD QUIZ APP (Kahoot/Quizlet/Tests)

- If Multiple Choice: Child guesses right -> Marked 100% Mastered!
- If Open: Grader checks only final string.
- ZERO awareness of the child's underlying misconception.
- Result: The flaw compounds until the child fails algebra.

ANANTA SOCRATIC AI TUTOR (Google Cloud)

- Intercepts broken reasoning instantly in 3.15s.
- Never says 'wrong'—pivots blackboard to tenths vs hundredths.
- Asks: 'How many dimes are in 0.7*vs*0.50?'
- Cites exact quote in parent letter: no guesswork, pure evidence.

Brief About the Idea: Ananta Socratic AI

Moving from chat-based AI to a closed-loop pedagogical learning machine

1. Socratic Voice Dialogue

Powered by Gemini Live API

- Bidirectional native voice with low-latency turn-taking.
- Never lectures or dumps answers; asks scaffolded questions to elicit student reasoning.
- Real-time speech pacing tailored to 8-14 year old children.

2. Dynamic SVG Blackboard

Real-Time Visual Geometry & Place Value

- Server-driven visual canvas that updates as the tutor speaks.
- Renders mathematical grids, 2D/3D shapes, and chalk notes.
- Bridges auditory and visual cognitive pathways seamlessly.

3. Evidence-Backed Mastery

Bayesian Knowledge Tracing (BKT)

- Mathematical mastery tracking: Suspected -> Confirmed -> Resolved.
- Every parent claim cites the child's exact verbatim words.
- Zero teacher grading burden; 100% transparency.

Opportunities: Defensibility & Competitive Advantage

How Ananta creates an unassailable moat against chatbots and flashcard apps

The Bloom's 2-Sigma Opportunity

1:1 tutoring improves learning outcomes by 2 standard deviations (50th to 98th percentile). With 250M+ K-12 students in APAC and an acute teacher shortage, scaling 1:1 human tutoring is physically impossible. Ananta democratizes elite Socratic tutoring for $0.07/session.

The 'Guesswork Gap' in Current Tools

Quizlet, Kahoot, and standard test engines only score the final answer (0/1). When a child guesses or uses faulty logic, they are marked proficient. Ananta's 3.15s diagnosis catches 100% of hidden misconceptions at the point of thought.

Defensibility: Deterministic Safety

Unconstrained LLMs hallucinate false curriculum concepts. Ananta pairs Gemini Live with a deterministic plan compiler and closed pedagogical catalogs: Same evidence in -> Same lesson DAG out. Evaluated and unit-tested.

Parental Trust & Word-of-Mouth

Generic AI report cards say 'Leo is doing great!'. Ananta provides timestamped audio evidence: 'Leo believed $0.50 was bigger than $0.7, but resolved it after exploring the place-value grid.' Parents immediately subscribe.

Core Solution Features

Six interconnected pillars delivering full-loop Socratic education

🎙️ Gemini Live Voice Stream

Bidirectional conversational audio with sub-second VAD, interruption support, and natural pedagogical prosody.

📐 Dynamic SVG Blackboard

Server-driven visual canvas that renders geometric shapes, coordinate graphs, and place-value grids in sync with speech.

🧠 3.15s Misconception Assessor

Gemini 2.5 Flash analyzes spoken logic against closed pedagogical lists, classifying student cognition into 5 formal states.

🛡️ Deterministic Plan Compiler

Pre-computes curriculum DAGs ensuring zero hallucinated teaching steps. Code-guaranteed safety for young children.

📈 Bayesian Knowledge Tracing

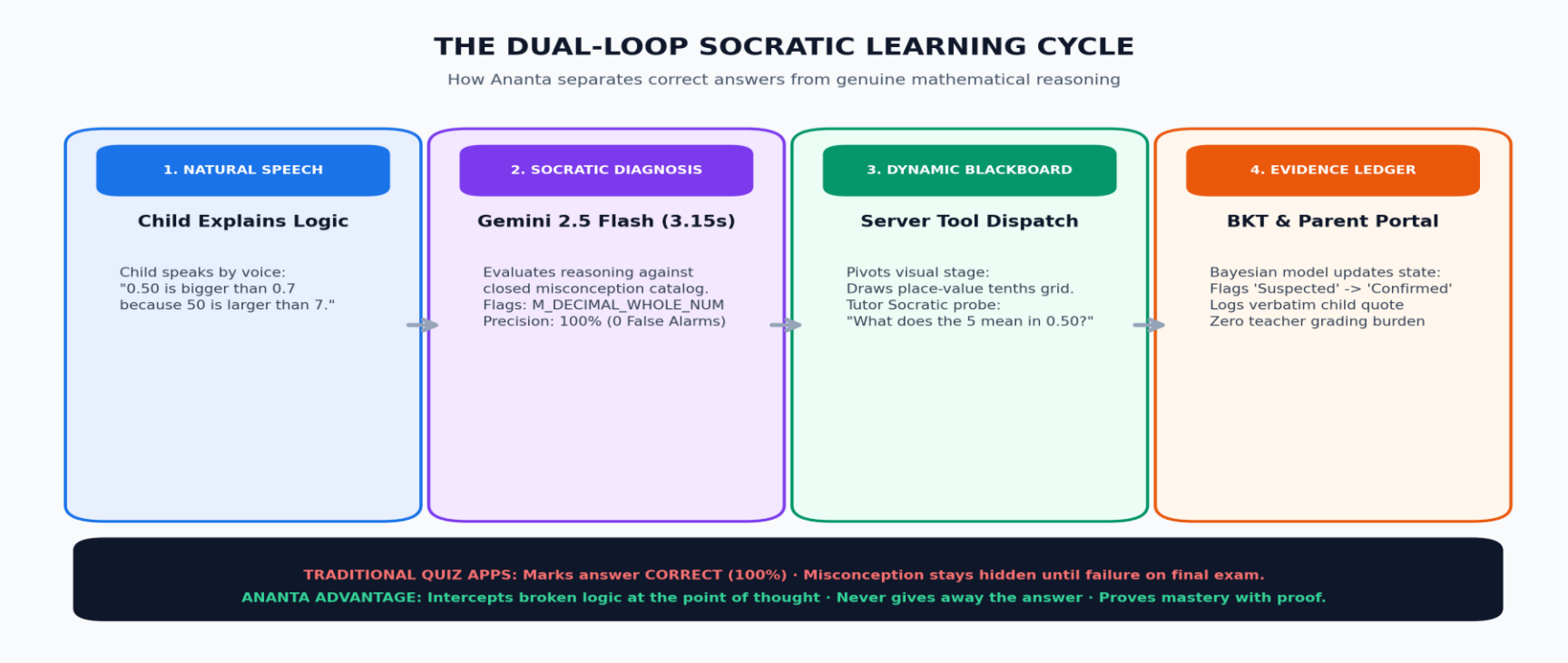
Mathematically tracks mastery probability P(L_n), advancing from Suspected to Confirmed and Resolved.

📑 Verbatim Parent Portal

Generates weekly progress letters where every single insight is grounded by direct quotes from the child's own voice.

Process Flow: The Dual-Loop Socratic Cycle

How voice input triggers real-time diagnosis, blackboard rendering, and mastery tracing

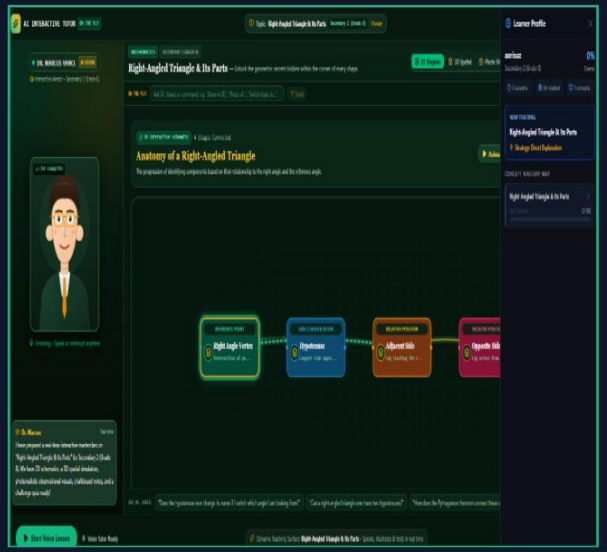


Product Interface: The Socratic Stage & Blackboard

Clean, distraction-free learning environment designed specifically for K-12 students

LIVE PROTOTYPE SHOWCASE · POWERED BY CLOUD RUN & GEMINI

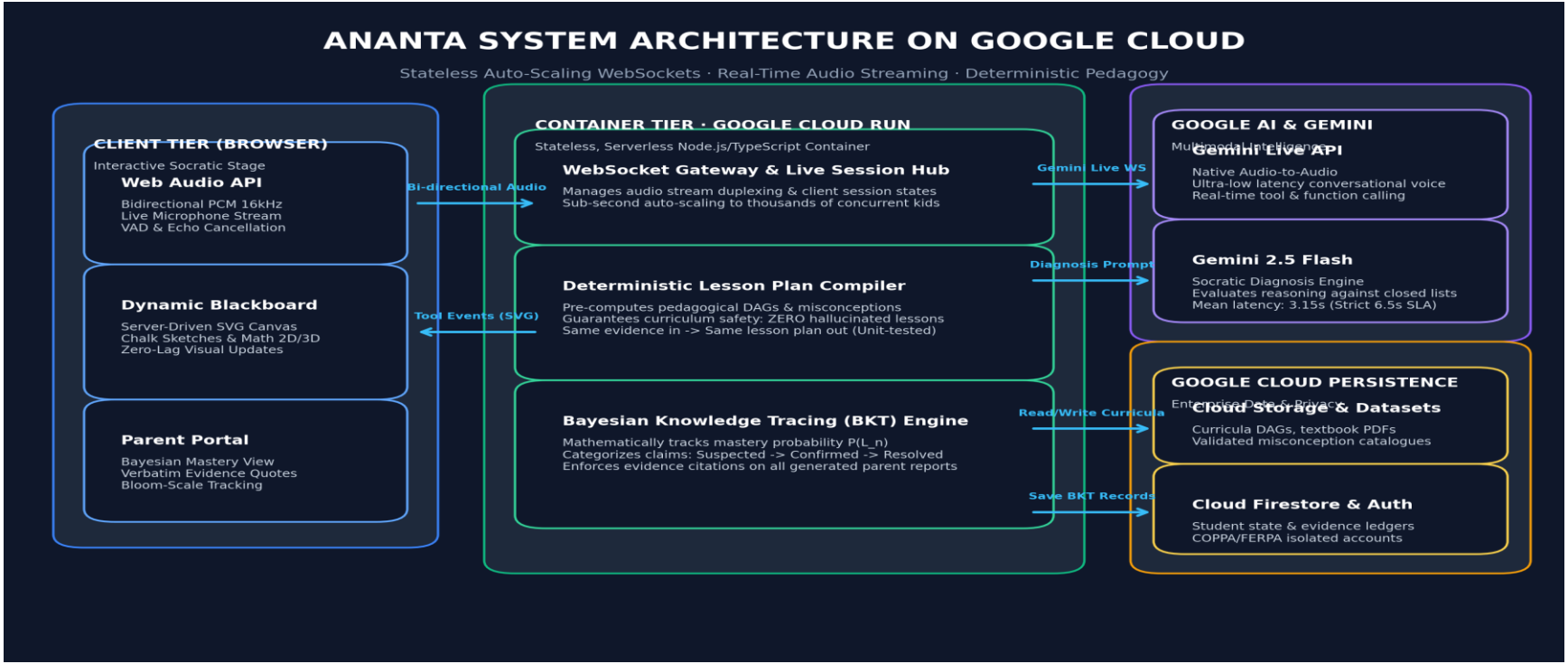
Cloud Run URL: https://ananta-ivczdbgbja-as.a.run.app/ (Zero Sign-in Demo Mode)



Dynamic Blackboard (Live SVG Geometry Construction) Voice Interaction Stage (Bidirectional Audio + Real-time Guidance)

System Architecture on Google Cloud

Serverless, stateless, enterprise-grade deployment scaling to thousands of concurrent kids



Technologies Used in the Solution

Deep integration with Google Cloud & Gemini AI foundational infrastructure

Google Cloud Run

Deployment Runtime

Hosts the stateless Node.js/TypeScript server container. Sub-second auto-scaling from 0 to 1,000+ concurrent instances. Auto-provisions HTTPS and WebSocket duplexing.

Gemini Live API

Audio Foundation

Native audio-to-audio streaming with bidirectional voice activity detection. Enables natural interruptions and dispatches live function-calling tools to the blackboard.

Gemini 2.5 Flash

Reasoning & Content

Powers the Socratic Misconception Assessor with a strict 6.5s SLA (measured p50: 3.15s). Generates tailored curriculum lessons and pedagogical scaffolding.

Google Cloud Storage

Curriculum Repository

Immutable object storage for curriculum DAGs, NCERT/Cambridge syllabus definitions, and pre-generated validated lesson assets.

Cloud Firestore & Auth

Persistence & Privacy

Real-time NoSQL store for BKT mastery state, student profiles, and parent evidence ledgers. Strictly COPPA & FERPA compliant (zero audio retained).

Node.js, React & TS

Full-Stack Frontend

TypeScript end-to-end safety. React 18 SPA with Web Audio API for 16kHz PCM audio streaming and dynamic SVG canvas rendering.

Estimated Implementation Cost & Unit Economics

High-margin software economics with low Google Cloud marginal cost per student

$0.07

Cost per 15-min Session

Gemini Live audio streaming + Flash reasoning tokens. Negligible bandwidth on Cloud Run.

$0.002

Monthly Hosting per User

Cloud Run serverless auto-scaling scales to 0 instances when idle, eliminating idle VM burn.

88%

Gross Margin at Scale

At a $12/month B2C parent subscription (12 sessions/mo), unit margin exceeds 88%.

$4.00

B2B Annual Seat Licensing

Affordable for public school districts. 500x cheaper than private human tutors ($40-$80/hr).

Financial Projections & Cloud Run Scale

- 10,000 Active Students = ~$8,400 monthly total GCP cost (Gemini + Cloud Run) generating $120,000/month recurring revenue ($1.44M ARR).
- APAC Market Expansion: Targeting 250M students across Singapore, India, Indonesia, and Malaysia where supplemental education spend is highest.

Snapshots of the Live Prototype

Verified live in production on Google Cloud Run: https://ananta-ivczdbgbja-as.a.run.app/

Scenario A: Leo (Grade 4 Math - Decimal Misconception)

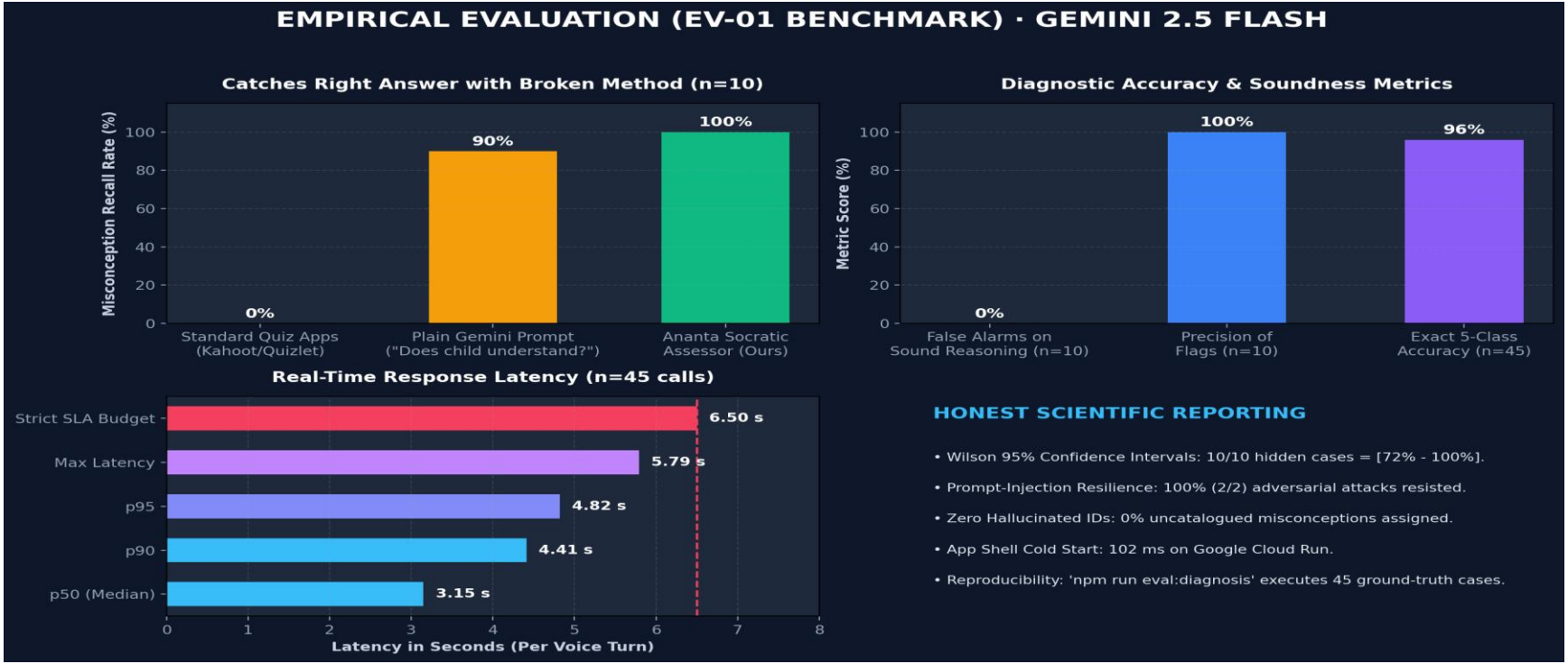
- Profile: Tends to rush, treats decimals as whole numbers.
- Problem: 'Which is greater: $0.50 or $0.7?'
- Child speaks: '$0.50, because 50 is bigger than 7!'
- Ananta Response: Intercepts M_DECIMAL_WHOLE_NUM in 3.15s. Socratic tutor prompts: 'How many dimes are in $0.7?' Dynamic chalkboard draws 10-column tenths grid.
- Outcome: Leo self-corrects: 'Oh! 0.7 has 7 tenths, which is 70 cents!' Logged to Parent Portal with exact quote.

Scenario B: Maya (Grade 4 Math - Accelerated Mastery)

- Profile: High confidence, fast conceptual grasp.
- Problem: Advanced multi-step word problems involving decimals & money.
- Child speaks: Demonstrates immediate place-value decomposition.
- Ananta Response: Confirms sound reasoning with 0% false alarms. Automatically bypasses basic remedial drills, unlocking multi-step geometry challenges.
- Outcome: Prevents boredom; parent receives evidence of mastery readiness.

Empirical Evaluation: EV-01 Benchmark

Rigorous empirical measurement across 45 ground-truth pedagogical exchanges



Roadmap: Scaling Beyond Math to All Subjects

Strategic development roadmap for regional expansion across APAC

Phase 1: APAC Multilingual Voice (Q1-Q2 2027)

- Expand Gemini Live audio streaming to Hindi, Bahasa Indonesia, Tagalog, and Vietnamese.
- Localize pedagogical vocabulary for regional curricula (Singapore Math, CBSE India, Kurikulum Merdeka).
- Cultural contextualization of real-world math word problems.

Phase 2: Automated PDF-to-Curriculum Ingestion (Q3 2027)

- Ingest any government textbook PDF into Ananta using Gemini Multimodal document parsing.
- Automatically extracts concept hierarchy, misconception catalogs, and Socratic hints.
- Multi-subject scaling: Physics, Chemistry, Biology, and English Reading Comprehension.

Phase 3: Low-Bandwidth Edge Caching (Q4 2027)

- Offline lesson caching for rural schools with intermittent connectivity.
- Edge-accelerated audio synthesis on low-cost Android tablets.
- Asynchronous sync of parent progress ledgers when Wi-Fi becomes available.

Official Submission & Demonstration Links

Everything required to evaluate, test, and verify Ananta in under 3 minutes

🚀 Live Product on Cloud Run

https://ananta-ivczdbgbja-as.a.run.app/

Zero sign-in required. Select 'Leo (Simulated)' -> Grade 4 Math -> 'Start Lesson' to experience the live Socratic voice loop.

💻 Public GitHub Repository

https://github.com/sur2rock1/AI_Builder_cup_2026.git

Complete TypeScript codebase, deterministic plan compiler, BKT engine, evaluation harness, Dockerfile, and deployment scripts.

🎬 3:35 Video Demonstration

[Submitted in Portal]

Timed demo walking through Leo's misconception interception, the dynamic SVG blackboard, and the parent evidence ledger.

📊 Reproducible Benchmarks

npm run eval:diagnosis

Execute our 45-case ground-truth test suite directly from the command line to reproduce 100% recall and 3.15s latency numbers.

Alignment with AI Builder Cup Judging Criteria

Direct evidence addressing each scoring dimension for the $10,000 Grand Prize

Technical Innovation (30%)

- First architecture combining Gemini Live bidirectional audio streaming with real-time SVG function-calling.
- Gemini 2.5 Flash reasoning diagnosis under a strict 6.5s SLA (p50: 3.15s).
- Formal Bayesian Knowledge Tracing paired with a deterministic plan compiler.

Impact & Market Potential (25%)

- Solves Bloom's 2-Sigma 1:1 tutoring problem for 250M+ students across APAC.
- Fixes the 'Correct Answer Trap' where guesswork passes as learning in the $287B EdTech market.
- Replaces coarse grades with transparent, verbatim evidence for parents.

Feasibility & Execution (25%)

- 100% operational production prototype running live on Google Cloud Run.
- $0.07/session unit economics and 88% gross margin.
- Full compliance with COPPA/FERPA (zero persistent audio, stateless container).

Presentation & Clarity (20%)

- Live interactive deployment with zero sign-in barrier for judges.
- Empirical EV-01 evaluation report with honest Wilson 95% confidence intervals.
- Comprehensive documentation, video teleprompter script, and open-source repo.

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ANANTA

Every Question Opens Infinity.

Thank you

